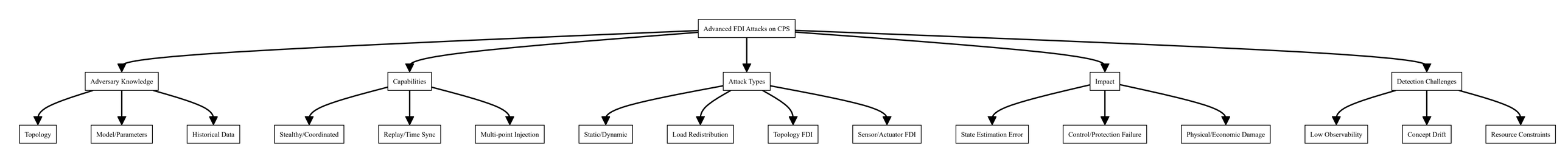
Advanced FDI Attacks on CPS
Adversary Knowledge
Topology
Model/Parameters
Historical Data
Capabilities
Stealthy/Coordinated
Replay/Time Sync
Multi-point Injection
Attack Types
Static/Dynamic
Load Redistribution
Topology FDI
Sensor/Actuator FDI
Impact
State Estimation Error
Control/Protection Failure
Physical/Economic Damage
Detection Challenges
Low Observability
Concept Drift
Resource Constraints